02LQD Specification and Simulation of Digital Systems

2023/01/23

1. Combinational circuit design (2 points)

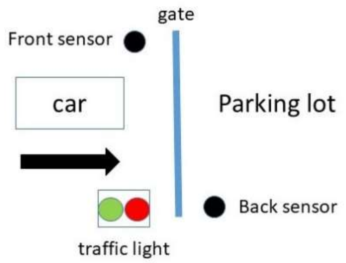
Design a combinational circuit that serves as a nonlinear LUT. It receives an 8-bit unsigned binary number LUTin and outputs an 8-bit unsigned binary number LUTout. The 2 nibbles of LUTout depend on the 2 nibbles of LUTin according to the following table:

Nibble #1 (bits 7:4)		Nibble #2 (bits 3:0)	
LUTin(7:4)	LUTout(7:4)	LUTin(3:0)	LUTout(3:0)
0000	0001	0000	1111
0001	1011	0001	0000
0010	1001	0010	1101
0011	1100	0011	0111
0100	1101	0100	1011
0101	0110	0101	1110
0110	1111	0110	0101
0111	0011	0111	1010
1000	1110	1000	1001
1001	1000	1001	0010
1010	0111	1010	1100
1011	0100	1011	0001
1100	1010	1100	0011
1101	0010	1101	0100
1110	0101	1110	1000
1111	0000	1111	0110

Design in VHDL a combinational circuit implementing the above specification according to the behavioral description style using the CASE WHEN statement (it is not necessary to explicitly write all input combinations, what matters is the overall structure of the description).

2. FSM design (12 points)

The entrance gate of a parking lot is controlled by a car parking system. Just in front of the gate there is a front sensor F and a traffic light (TL). Just after the gate there is a back sensor B. An external device (not to be designed) checks the code of the car and generates an ok input to the system.



The traffic light may be off ($TL=l_1l_0=00$) or flashes green ($TL=l_1l_0=01$) or red ($TL=l_1l_0=10$). The front sensor either detects a car willing to enter the parking lot (F=1) or no car is willing to enter (F=0). The back sensor detects whether the car has crossed the gate and is in the parking lot (B=1) or still crossing (B=0). The car never backs up.

If no car is trying to enter the parking lot (F=0), the system keeps the traffic light off and the gate is closed. If the front sensor detects a vehicle (F=1), the system turns on the red light as long as the code is wrong (ok=0). As soon as a correct code is detected (ok=1), the systems flashes the green light and opens the gate. While the car is crossing the gate, the back sensor is B=0 and no other car willing to enter is allowed to (the traffic light is red). As soon as the car has crossed the gate, B=1 and operations continue depending on the presence/absence of a car willing to enter.

Design a **Mealy** FSM with F, B and ok as inputs and TL as output.

1. Draw the state diagram (**4 points**)
2. Design the FSM resorting to the VHDL behavioural description style with 2 processes (**4 points**).
3. Design the FSM resorting in Verilog with 2 always blocks (**4 points**).

3. FSM-D design (16 points)

In arithmetic the **least common multiple** of two positive integers *a* and *b*, usually denoted by **lcm(*a*, *b*)**, is the smallest positive integer that is divisible by both *a* and *b*. Since division of integers by zero is undefined, this definition has meaning only if *a* and *b* are both different from zero.

Examples:

- a = 0, b ≠ 0: lcm(0, b) = 0
- a ≠ 0, b = 0: lcm(a, 0) = 0
- a = 7, b = 7: lcm(7, 7) = 7
- a = 12, b = 18: lcm(12, 18) = 36

A simple algorithm to compute lcm(a,b) is:

```
m=a
n=b
while (m != n) {
  if (m < n)
    m=m+a
  else
    n=n+b
}
lcm=m
```

Design a FSM-D to compute the lcm of 2 pure binary numbers a and b on 8 bits implementing the above algorithm. Computation starts when a start signal is asserted for one clock cycle. When computation is over the ready signal is asserted for one clock cycle. Reset is asynchronous. Values for lcm are available at each computation step, but they make sense only when ready is 1.

Steps:

4. Define the appropriate size for the lcm output (**2 points**)

5. Draw the HLSM state diagram and write the VHDL code to capture the HLSM behavior (**6 points**)

6. Design a FSM-D:

- create 2 processes for behavioural datapath description from the HLSM state diagram (4 **points**)
- create 2 processes for behavioural controller description from the HLSM state diagram (**4 points**).

Exam rules:

- upload on Portale della didattica in section Elaborati a .zip archive including:
  - running VHDL code and testbench
  - short report (max 3 pages) describing the solution highlighting differences w.r.t. the solution handed in at the written exam

  no later than Thursday January 26 11.59 pm
- format of .zip archive name surname_ID (example Rossi_244123)
- remember: no upload ⇒ no correction ⇒ no oral exam
- once the ranking is published on Portale della didattica in section Materiale/Results, drop an e-mail to paolo.camurati@polito.it to agree on a date for the oral exam.